Source-to-Target Mapping & Reconciliation Work Instruction

Document ID	**WI-DATA-022**
Revision	1.0
Status	ILLUSTRATIVE — NOT APPROVED FOR OPERATIONS
Audience	Data Engineering, Business Data Owners, QA/UAT Testers
Information classification	Synthetic portfolio exercise; no employer or customer data

Scenario: A fictional business consolidates five legacy datasets into a new ERP staging environment. All systems, transactions, personnel and results in this document are synthetic.

1. When to use

Use this work instruction to review incoming synthetic data extracts, map field values into AtlasERP staging, verify transformation results, and route inconsistencies to authorized data owners. Do not treat a passing mapping validation as business release approval.

2. Data sources and authoritative owners

Legacy source	Synthetic dataset	Owner	Target entity
LEG-A SupplierPortal	Supplier identity and approval	Procurement	Vendor
LEG-B MaterialMaster	Material, supplier association, storage	QA/Manufacturing	Material
LEG-C Orders360	Orders and ship-to references	Order-to-Cash	PurchaseOrder
LEG-D RegTrack	Market eligibility and registration dates	Regulatory	ProductMarket
LEG-E FinanceLedger	Currency, terms and price	Finance	PriceCondition

3. Field-level mapping specification

Mapping ID	Source -> target	Transformation/validation	Expected exception
MAP-01	LEG-A.vendor_code -> Vendor.vendor_id	Trim; uppercase; require VND-####	Format error
MAP-02	LEG-A.approval_status -> Vendor.status	Map APPROVED/ACTIVE to Active only with approval evidence	Missing approval
MAP-03	LEG-B.material_sku -> Material.material_id	Trim; uppercase; preserve original key in lineage	Duplicate SKU
MAP-04	LEG-B.vendor_code -> Material.vendor_id	Foreign-key lookup against accepted Vendor	Orphan vendor
MAP-05	LEG-C.order_amount -> PurchaseOrder.amount_usd	Parse decimal; reject unknown currency	Invalid currency
MAP-06	LEG-D.market_code -> ProductMarket.market	Whitelist US,CA,EU and record original value	Unknown market
MAP-07	LEG-E.payment_terms -> PriceCondition.payment_terms	Translate NET30/NET45; route unrecognized terms	Unsupported term
MAP-08	All.source_id ->	Retain exactly as source;	Missing lineage

	*.source_record_id	never generate a replacement

4. Step-by-step work procedure

- Step 1 — Obtain extracts, checksum or file inventory, extraction timestamp, schema version, and source record counts from the authorized source owner.
- Step 2 — Perform profiling checks for missing values, duplicates, illegal characters, inconsistent codes, unexpectedly high null rates, and broken associations.
- Step 3 — Compare each transformed field against its mapping ID. Preserve both source and target values in test evidence; never modify the source fixture to force a pass.
- Step 4 — Run pre-load referential checks; reject material rows linked to unapproved or nonexistent supplier IDs. A rejected row must be visible in the exception log.
- Step 5 — After a synthetic test transfer, reconcile record counts by entity and status; compare high-risk fields in 100% of designated critical test cases.
- Step 6 — Send unresolved vendor issues to Procurement, material issues to QA/Manufacturing, market codes to Regulatory, and monetary issues to Finance.
- Step 7 — Retest a corrected extract against the same expected results. Record environment, mapping version, reviewer, timestamp, and pass/fail outcome.
- Step 8 — Recommend acceptance only when designated blocking issues have evidence of closure; the business owner makes the actual acceptance decision.

5. Reconciliation controls

Control	Procedure	Tolerance
Record count	Compare extract totals + accepted + rejected + quarantined	All rows accounted for
Critical attributes	Compare sample source, mapped target and authority record	Zero critical mismatches
Value totals	Compare source and target USD totals by approved currency	0.00 difference on synthetic test set
Foreign key	Verify every material has valid accepted vendor	Zero unresolved orphans
Revision lineage	Link mapping version to each test execution	100% executions linked

6. Worked discrepancy example

Example: supplier VND-0042 is APPROVED in the procurement export but has a missing approval-reference field. A naive transform would mark the vendor Active. Under MAP-02, the row is rejected as EXC-003, Procurement is asked for approval evidence, and QA/UAT verifies the row is not selectable for a new material while unresolved.

The correction pathway is to obtain a valid controlled approval record or keep the supplier blocked. The portfolio does not assert that a real vendor was approved.

7. Common failure modes and recovery

- Whitespace or mixed-case IDs: Normalize only where the signed mapping explicitly permits it; preserve source value.
- Duplicate legacy keys: Do not overwrite prior values; create a duplicate cluster and request an authoritative master.
- Missing values in critical fields: Reject or quarantine; do not substitute dummy values.

- Failed load: Preserve the last accepted snapshot, isolate failed batch, review logs and retest after correction.

8. Evidence package and handoff

Attach synthetic input/output samples, mapping ID, transform version, file counts, comparison results, relevant exceptions, and owner decisions. Retain as a UAT evidence bundle, not as proof of live production validation.

REGULATORY / TECHNICAL REVIEW ADDENDUM — November 2026

Independent technical writing portfolio enhancement. Additional validation gates and examples are illustrative, not authorized production instructions. Regulated applicability varies by product, process, record and organization.

Field mapping and deterministic transformation rules

Example field VENDOR_ID: Legacy A SUPPLIER_CODE / Legacy B SUPP_ID -> ERP BP_VENDOR_ID. Normalize whitespace, preserve meaningful leading zeros, reject duplicate legal IDs, and match approved cross-reference table revision MAP-004. Missing identifiers enter an exception queue rather than being guessed.

Example MATERIAL_ID: derive from an approved controlled materials crosswalk, not free-text name similarity. Map manufacturer and purchasing supplier separately when the source distinguishes them; retain original lot and supplier references in a read-only reconciliation audit file.

Example SPEC_REV: current approved specification wins only through explicit authorized change control; an obsolete source revision must be flagged, never silently replaced in the evidence trail.

Reconciliation framework and evidence

Pre-load: obtain authorized extracts, record counts, file hashes or equivalent integrity controls, data dictionary revision, timestamp, extract owner and retention location. Transformation: document mapping version, run ID, rejected records and reference data. Post-load: compare expected/actual counts, required field population, supplier status, material associations, referential integrity, and exception totals.

Example reconciliation: Source count 1,000; target loaded 985; quarantined 15 with exception IDs. Count reconciliation balances only if source = accepted + rejected with no unexplained loss and known deduplication rules. Numbers are fictional illustrative arithmetic, not tested results.

Security: use test identities, protected transfer channels and role-based access; record who can approve mapping changes. Do not store real vendor, banking or customer personal data in the GitHub evidence examples.

Relevant guidance boundary

Use ISO/IEC/IEEE 26514:2022 as guidance for the *user documentation*, not as a software validation standard. If a deployment supports FDA-regulated records, assess 21 CFR Part 11 with the predicate-rule analysis and company validation processes; do not promise universal compliance.

Document-specific reference basis and status

ISO/IEC/IEEE 26514:2022 information for software users (full URL in portfolio review report)

IEC/IEEE 82079-1:2019 instructions for use (full URL in portfolio review report)

FDA Part 11 scope and application guidance (full URL in portfolio review report)

W3C WCAG 2.2 color use guidance (full URL in portfolio review report)

Status: technical/editorial review against publicly available references completed at a portfolio-example level; no licensed-standard full-text conformance audit, live environment testing, site QA approval or legal certification performed.

Appendix: November 2026 standards and documentation applicability review

PUBLIC SYNTHETIC PORTFOLIO SAMPLE — This appendix records source-checking and editorial scope boundaries. It does not make the fictional organization regulated, certify a system, or replace actual company policies and competent quality/legal review.

ERP / UAT regulatory applicability and software test-evidence review

- Confirm each source-to-target mapping has an owner, approved transformation rule, exception disposition and independent tie-out. Synthetic test failures must remain OPEN until a documented rerun passes.
- Separate functional UAT from regulated computerized-system validation: business acceptance is not evidence of 21 CFR Part 11 compliance or FDA-required validation.
- 21 CFR Part 11 may become relevant only if the mapped environment maintains qualifying electronic records/signatures within predicate-rule scope; evaluate access, audit trail, record retention and copies when applicable.
- For medical-device production/QMS software, FDA's February 2026 Computer Software Assurance final guidance provides risk-based recommendations and QMSR became effective February 2, 2026; do not impose these onto a general commercial SAP migration.
- NIST access control, change management and audit logging offer voluntary security reference points for privileged ERP access, defect fix approvals and evidence retention.
- Technical writing QA: include acceptance criteria, reproducible steps, observed versus expected result, severity basis, defect owner, approval and version; never infer that an unresolved hypothetical defect passed.

Reference links and recheck notes

https://www.ecfr.gov/current/title-21/chapter-I/subchapter-A/part-11

https://www.fda.gov/regulatory-information/search-fda-guidance-documents/computer-software-assurance-production-and-quality-management-system-software

https://csrc.nist.gov/pubs/sp/800/53/r5/upd1/final

Verified against published authoritative source pages as of November 2, 2026. No proprietary standards were audited clause-by-clause. Actual application must be assessed based on product, system use, region, and record type. Proposed procedures have not been executed in a production environment.

Independent second-pass document review | November 2, 2026

REVIEWED PORTFOLIO DEMONSTRATION — this appendix records a document-specific editorial and source-scope check. The main procedure and forms are fictional and not approved for production, executed testing or regulated use. Original source versions are retained as editable files in this portfolio.

A. Document-specific findings and checks

Review objective	2026 reviewer correction / expectation	Verification evidence
Mapping provenance	For each of five legacy sources, define field definitions, transform, null handling, master identifier and ownership.	Synthetic mapping IDs and row-level examples reconcile to register.
Reconciliation acceptance	Compare counts, required-field completeness, exception lists and duplicate supplier conflicts.	Discrepant data must not be accepted merely because total row counts match.
Change traceability	Track mapping rule revision, defect, source record, test ID and retest evidence.	Keep prior input and previous rule rather than rewriting a failed historical test.

B. Applicable regulatory / guidance boundaries

Review objective	2026 reviewer correction / expectation	Verification evidence
Scope	Fictional commercial supplier master/ERP migration, not a validated medical-device QMS or clinical EDC.	Do not label SAP-like mock UAT as an FDA validation or actual system access test.
Evidence	Requirements -> source mapping -> expected/actual result -> defect -> retest -> acceptance owner.	Use synthetic IDs, revisioned mappings and traceable test executions.
Conditional FDA applicability	Part 11 if predicate electronic records; 2026 QMSR and device CSA only if used for in-scope device production/QMS.	A business test pass never independently proves GxP compliance or authorized production deployment.

C. Evidence and sign-off boundaries

Status: Editorial second-pass reviewed; technical procedure not executed; company-specific SOP approval not obtained.

All tests, screenshots, users, suppliers, dates, IDs, incident observations, performance metrics and signatures in the demonstration are synthetic.

Next practitioner action: confirm source applicability for the actual jurisdiction and product; conduct SME/user testing and accessibility review; update controlled revision and approval fields.

No color, watermark, revision number, PDF export or reference link by itself demonstrates regulatory compliance.

D. Authoritative primary-source references

FDA 21 CFR Part 11 scope: https://www.fda.gov/regulatory-information/search-fda-guidance-documents/part-11-electronic-records-electronic-signatures-scope-and-application

FDA device QMSR Feb 2026: https://www.fda.gov/medical-devices/postmarket-requirements-devices/quality-management-system-regulation-qmsr

FDA device CSA Feb 2026: https://www.fda.gov/regulatory-information/search-fda-guidance-documents/computer-software-assurance-production-and-quality-management-system-software

US Access Board Section 508: https://www.access-board.gov/ict/about/